

MD ANWAR HOSSAIN

Quantitative Ecology | Artificial Intelligence | Computational Modelling | Environmental Change

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Melbourne, VIC, Australia | Work rights: Australian Citizen

SUMMARY

Quantitative Ecologist with qualifications in artificial intelligence, big data analysis, species distribution modelling, and high-performance computing (HPC). Experienced in integrating data from various sources to deliver actionable insights. Skilled in R and Python, with developing proficiency in Julia. Published in high-impact journals (250+ citations) and presented at international conferences. Keen to apply advanced computational approaches, and AI techniques to solve innovative problems.

TECHNICAL SKILLS

- Programming & Modelling: R, Python, Julia (developing)
- Analytical Approaches: Applied Natural Language Processing, Neural Network and Deep Learning, Data Warehousing and Mining, Advanced Object-Oriented Programming, Species Distribution Models (SDM), Generalized Dissimilarity Models (GDM)
- Computing: High-Performance Computing (HPC), Linux, SLURM workload manager
- Data & Spatial Analysis: Big Data handling, Remote Sensing, GIS, Computational Modelling, Data Visualization
- Research & Collaboration: Interdisciplinary teamwork, Stakeholder engagement, Scientific writing & presentations

RESEARCH EXPERIENCE

PhD Research – Quantitative Ecology

University of Melbourne, Australia | 2018–2023

- Built a national database of historic Australian grasshopper occurrences from handwritten survey notebooks.
- Transcribed, georeferenced (Google Earth, topographic maps), and curated records to detect surveyors and spatial biases.
- Integrated soil, vegetation, and climate layers into SDMs and GDMs to analyze diversity patterns.
- Published 4 peer-reviewed papers during PhD candidature.

MPhil Research – Quantitative Ecology

University of Melbourne, Australia | 2016–2018

- Collated life-history traits for 574 freshwater crayfish species and integrated with Global Circulation Models (GCMs) in trait-based climate change vulnerability assessments (TVA).
- Identified global hotspots and vulnerable species; evaluated alignment of TVA outcomes with IUCN Red List, ESA, and EPBC Act protocols; analyzed GCM uncertainties in an HPC environment using Linux and SLURM.
- Awarded 90% marks (High Distinction) in thesis; 2 publications in high-impact journals.

Teaching & Research Experience

Practical Demonstrator (Casual)

University of Melbourne, 2019–2022

- Led undergraduate (3rd yr) and Masters practical sessions in Environmental Risk Assessment courses.
- Responded to student queries and provided feedback on coursework.
- Trained and mentored undergraduate and postgraduate students in ecological methods and data analysis.
- Marked assignments and exam papers; supervised online exams.

Research Assistant (Casual)

University of Melbourne, 2021 – 2023

- Maintained grasshopper colonies for experimental projects, supporting research continuity.
- Supported data collection and experimental setup for ecological modelling projects.

Research Assistant - Zoology

University of Dhaka, Bangladesh, 2012–2015

- Conducted faunal surveys, managed data collection, and produced ecological research reports.
- Appointed as a Team Leader for multiple surveys; coordinated field logistics, supervised staff, and delivered final reports on schedule.

CONSULTANCY & COMMUNITY CONTRIBUTIONS

- Conservation Consultant | WildTeam, Bangladesh | 2024–2025
- Reviewer for Wiley and Elsevier journals.
- Regular blood donor.

EDUCATION

- Graduate Certificate in Artificial Intelligence, Victoria University (June, 2026)
- PhD in Quantitative Ecology, University of Melbourne, 2018–2023
- MPhil in Quantitative Ecology, University of Melbourne, 2016–2018 — Thesis mark: 90%
- MSc in Zoology (Wildlife), University of Dhaka, 2011–2012 — GPA: 3.93/4.0
- BSc (Hons) in Zoology, University of Dhaka, 2007–2011 — CGPA: 3.83/4.0

AWARDS & CERTIFICATIONS

- **Gold Medal** – University of Dhaka (2012, top of MSc cohort)
- **Dean's Award** – University of Dhaka (2011, academic excellence in BSc Honours)
- **Certifications (WildTeam UK)**: Project Planning, Project Management, Monitoring & Evaluation for Wildlife Conservation

SELECTED PUBLICATIONS (see full list here: <http://go.unimelb.edu.au/b6qr>)

- **Hossain, M.A.**, Lahoz-Monfort, & Kearney, M.R. (2024). Leveraging historical field notebooks to uncover continental-scale patterns in the diversity of Australian grasshoppers. *Insect Conservation and Diversity*, 17(3): 488-500.
- **Hossain, M.A.**, Lahoz-Monfort, & Kearney, M.R. (2023). Developing a database of Australian grasshopper occurrences from historic field survey notebooks spanning 54 years (Orthoptera: Acrididae, Morabidae, Pyrgomorphidae, Tetrigidae). *Austral Entomology*, 62(1): 64-76.
- **Hossain, M.A.**, et al. (2019) Assessing the impacts of uncertainty in climate change vulnerability assessments. *Diversity and Distributions*, 25(8): 1234-1245.
- **Hossain, M.A.**, et al. (2018) Assessing the vulnerability of freshwater crayfish to climate change. *Diversity and Distributions*, 24(12): 1830-1843.

SELECTED CONFERENCE PRESENTATIONS

- *Oral presentation (online)* - International Biogeography Society, Vancouver, 2022
- *Oral presentation* - Australian Entomological Society, Brisbane, 2019
- *Oral presentation* - International Congress for Conservation Biology, Colombia, 2017
- *Invited oral presentation* - Joint Meeting of Ichthyologists and Herpetologists, New Orleans, 2016
- *Poster presentation* - Student Conference on Conservation Science, Beijing, 2014

REFERENCES

- Prof. Michael Kearney, School of BioSciences, University of Melbourne | m.kearney@unimelb.edu.au
- Dr José Lahoz-Monfort, Senior Research Fellow, University of Melbourne | jose.lahoz@unimelb.edu.au